Stage 3

User Story: Loan Application Management System

As a user, I want to apply for a loan after successful login, and as an admin, I want to review, approve, reject, or request additional information for loan applications.

1. User Side Task: Loan Application

After a user successfully logs in, they should be able to apply for a loan using the console-based application.

1.1 Loan Application

The loan application should collect the following details:

- Loan Amount (loanAmount)
- Loan Type (loanType) – Personal, Home, Education, Car, etc.
- Loan Duration (loanDuration) – Duration of the loan in months or years.
- Interest Rate (interestRate) – This can be auto-filled based on loan type.
- Annual Income (annualIncome) – User's income for determining eligibility.
- Purpose of Loan (purpose) – A brief description of why the loan is needed.
- Status (status) – Automatically set to Pending when the user submits the loan.
- User ID (userId) – This will link the loan application to the user.

1.2 Functionality

Form Validation

- Ensure all fields are required except status, which defaults to Pending.
- Validate that the loan amount and annual income are positive numbers.
- Loan duration must be a valid number greater than 0.
- Required inputs should be accepted using the **Scanner class**.

Loan Application Submission

- After submission, the loan application should be saved to the MySQL database using JDBC.
- Loan status should be saved as Pending.
- Notify the user that their loan application has been submitted successfully.

2. Admin Side Task: Loan Application Review

Once a user applies for a loan, the admin should have the ability to review and take appropriate actions on the application.

2.1 Loan Review Menu

The admin should be able to view loan applications with the following details:

- Loan ID
- User's Full Name
- Loan Amount
- Loan Type
- Application Date
- Status (Pending, Approved, Rejected, Needs More Info)

Admin should be able to:

1. View Loan Application Details.
2. Approve or Reject the loan.

2.2 Actions Admin Can Perform on Loan Application

1. View Loan Details

Admin should be able to view the full details, including:

- User's Name
- Email
- Mobile Number
- Loan Amount
- Loan Type
- Loan Duration
- Annual Income

- Purpose
- Application Date
- Status

2. Approve Loan

- Admin can approve the loan if everything is in order.
- Change the loan status to Approved.
- Notify the user through email.

3. Reject Loan

- Admin can reject the loan if the application does not meet the criteria.
- Change the loan status to Rejected.
- Provide a reason for rejection.
- Send the rejection reason to the user through email.

3. Email Notification System

After the admin takes any action on the loan, an **actual email should be sent to the user** with the appropriate details.

Email Notifications

1. Loan Approved

Subject: Your Loan Application has been Approved!

Body:

Dear [User], your loan application for [loan amount] has been approved. Please check your dashboard for further details.

2. Loan Rejected

Subject: Your Loan Application has been Rejected

Body:

Dear [User], we regret to inform you that your loan application has been rejected. Reason: [reason]. Please contact support for more details.

4. Loan Application Flow

User Applies for Loan

- User fills in the loan details using the **Scanner class** and submits the application.
- Loan status is Pending by default.
- Application is saved to the MySQL database using JDBC.
- User is notified that their loan application has been submitted successfully.

Admin Reviews Loan

- Admin logs in and sees loan applications of users, if any.
- Admin views pending loan applications and takes the appropriate action.
- Approves loan → Updates status to Approved → Sends an email to the user.
- Rejects loan → Updates status to Rejected → Sends an email to the user with the reason.

The application should be **console-based only**. No frontend, Spring Boot, Spring MVC, or Spring Security is required.

All database operations should be performed using **JDBC**.